

FINARTS HW 3 — Market Model Regression

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1 Overview

Model:

$$R_i = \beta_0 + \beta_1 \cdot R_m + u$$

Selected stocks (5 sectors):

Stock	RIC	Sector	Company
BPI	BPI.PS	Banks	Bank of the Philippine Islands
Century Pacific	CNPF.PS	Food Producers	Century Pacific Food
Meralco	MER.PS	Electricity	Manila Electric (Meralco)
ABS-CBN	ABS.PS	Media	ABS-CBN
PLDT	TEL.PS	Telecommunications	PLDT

2 Data Loading

```
library(dplyr)
library(tidyr)
library(readxl)
library(broom)

set.seed(42L)

company_returns <- read_excel("data/raw/Company>Returns.xlsx")
psei_return      <- read_excel("data/raw/PSEi_Return.xlsx")
company_info     <- read_excel("data/raw/Company Name.xlsx")

selected_rics <- c("BPI.PS", "CNPF.PS", "MER.PS", "ABS.PS", "TEL.PS")

stock_meta <- company_info |>
  filter(RIC %in% selected_rics) |>
  select(RIC, Name, Sector, Full_Name = `Full Name`)

knitr::kable(stock_meta, caption = "Selected Stocks Metadata")
```

Table 2: Selected Stocks Metadata

RIC	Name	Sector	Full_Name
BPI.PS	BANK OF THE PHILP.ISLE.	Banks	Bank of the Philippines Islands

RIC	Name	Sector	Full_Name
MER.PS	MANILA ELECTRIC	Electricity	Manila Electric
TEL.PS	PLDT	Telecommunications Service Providers	PLDT
CNPF.PS	CENTURY PACIFIC FOOD	Food Producers	Century Pacific Food
ABS.PS	ABS-CBN	Media	ABS-CBN

3 PART 1: Regression Analysis

Instruction: Select 5 stocks from 5 different sectors. Run a regression of the returns of each stock against the market returns.

$$R_i = \beta_0 + \beta_1 \cdot R_m + u$$

3.1 Question 1 (2 pts)

Create a dataframe for your regressions. It must contain the 5 stock returns, PSEi return, and date.

```
psei <- psei_return |>
  mutate(Date = as.Date(Date)) |>
  rename(PSEi_Return = PSEi_Return) |>
  distinct(Date, .keep_all = TRUE)

stock_returns <- company_returns |>
  filter(RIC %in% selected_rics) |>
  rename(Return = Total_Return, Date = Date) |>
  mutate(Date = as.Date(Date))

reg_data_long <- stock_returns |>
  inner_join(psei, by = "Date")

reg_data_wide <- reg_data_long |>
  select(Date, RIC, Return, PSEi_Return) |>
  pivot_wider(
    id_cols = c(Date, PSEi_Return),
    names_from = RIC,
    values_from = Return,
    names_prefix = ""
```

```

) |>
mutate(across(all_of(selected_rics), as.numeric)) |>
arrange(desc(Date))

# Displaying the first 5 rows of the generated regression dataframe
knitr::kable(head(reg_data_wide, 5), digits = 4,
              caption = "Preview of the Regression Dataframe")

```

Table 3: Preview of the Regression Dataframe

Date	PSEi_Return	BPI.PS	MER.PS	TEL.PS	CNPF.PS	ABS.PS
2025-12-29	1.1823	-2.9264	-0.1739	0.0000	0.2571	-1.6355
2025-12-26	-1.4563	-0.1669	-0.8621	0.0000	-1.1436	4.3902
2025-12-23	0.4878	0.4191	-0.8547	0.0000	0.8974	-4.2056
2025-12-22	0.4902	2.8448	1.7391	0.6390	0.1284	13.8298
2025-12-19	0.4926	-2.2746	-2.8716	-1.4173	-2.1357	5.0279

3.2 Question 2 (2 pts each, 10 pts total)

Run the regressions and report your estimated parameters.

```

stock_names <- list(
  "BPI.PS" = "Bank of the Philippine Islands",
  "CNPF.PS" = "Century Pacific Food",
  "MER.PS" = "Manila Electric (Meralco)",
  "ABS.PS" = "ABS-CBN",
  "TEL.PS" = "PLDT"
)

stock_sectors <- list(
  "BPI.PS" = "Banks",
  "CNPF.PS" = "Food Producers",
  "MER.PS" = "Electricity",
  "ABS.PS" = "Media",
  "TEL.PS" = "Telecommunications Service Providers"
)

sig_stars <- function(p_value) {
  if (is.na(p_value)) return("")
  if (p_value < 0.01) return("***")
  if (p_value < 0.05) return("**")
  if (p_value < 0.10) return("*")
  return("")
}

```

```

}

regression_results <- list()

for (ric in selected_rics) {
  stock_data <- reg_data_long |> filter(RIC == ric)
  model <- lm(Return ~ PSEi_Return, data = stock_data)
  regression_results[[ric]] <- model

  coef_summary <- summary(model)$coefficients
  b0_est <- coef_summary[1, 1]; b0_pval <- coef_summary[1, 4]
  b1_est <- coef_summary[2, 1]; b1_pval <- coef_summary[2, 4]

  cat("STOCK", which(selected_rics == ric), "\n")
  cat("RIC:", ric, "\n")
  cat("Company Name:", stock_names[[ric]], "\n")
  cat("Sector:", stock_sectors[[ric]], "\n")
  cat("Estimated parameters:\n")
  cat("B0:", sprintf("%.6f", b0_est), sig_stars(b0_pval), "\n")
  cat("B1:", sprintf("%.6f", b1_est), sig_stars(b1_pval), "\n\n")
}

```

```

STOCK 1
RIC: BPI.PS
Company Name: Bank of the Philippine Islands
Sector: Banks
Estimated parameters:
B0: 0.062712
B1: 0.014535

```

```

STOCK 2
RIC: CNPF.PS
Company Name: Century Pacific Food
Sector: Food Producers
Estimated parameters:
B0: 0.087240
B1: -0.029359

```

```

STOCK 3
RIC: MER.PS
Company Name: Manila Electric (Meralco)
Sector: Electricity
Estimated parameters:
B0: 0.120955 *
B1: 0.033800

```

```

STOCK 4

```

RIC: ABS.PS
Company Name: ABS-CBN
Sector: Media
Estimated parameters:
B0: -0.025594
B1: 0.023995

STOCK 5
RIC: TEL.PS
Company Name: PLDT
Sector: Telecommunications Service Providers
Estimated parameters:
B0: 0.032580
B1: 0.065881 *

(Note on significance: ** $p < 0.01$, * $p < 0.05$, $p < 0.10$)*

3.3 Question 3 (5 pts)

Discuss your findings. Discuss the results for each stock, and compare the estimated parameters for each of the stocks.

The empirical estimation of the market model across the five selected equities reveals a pervasive lack of statistical significance at the standard 5% threshold ($p > 0.05$ for all β_1 coefficients). For the timeframe evaluated (2023–2025), the systematic market movements dictated by the PSEi failed to meaningfully explain the daily variance of these specific stocks, a finding corroborated by near-zero R^2 values across the panel. Consequently, the daily returns of these selected equities appear to be driven predominantly by idiosyncratic, company-specific factors rather than broader macroeconomic market trends.

Despite the overarching statistical insignificance, examining the individual slope coefficients provides insight into the directional sensitivities of the equities. PLDT (TEL.PS) registers the highest market sensitivity among the cohort with a beta of 0.0659, emerging as the only stock to approach marginal significance at the 10% level ($p = 0.083$). This suggests it tracks the broader index slightly more closely than its peers. Similarly, Manila Electric (MER.PS) yields a positive beta (0.0338) coupled with a marginally significant positive alpha ($\beta_0 = 0.1210$, $p < 0.10$), hinting at slight positive excess returns generated independently of the market. Both the Bank of the Philippine Islands (BPI.PS) and ABS-CBN (ABS.PS) exhibit very weak positive betas (0.0145 and 0.0240, respectively) alongside high p-values, confirming that market fluctuations barely permeate their return profiles.

In contrast to the rest of the cohort, Century Pacific Food (CNP.F.PS) stands out as the only stock yielding a negative beta value. Specifically, the estimated parameters indicate that a 1% increase in the PSEi leads to a 0.029% decrease in Century Pacific's return. This inverse dynamic suggests a defensive, counter-cyclical characteristic inherent to the food producer, even though the relationship itself ultimately fails to achieve statistical significance ($p = 0.539$).

4 PART 2: Fitted Values and Residuals

4.1 Question 1 (5 pts)

Compute the fitted values of the 5 stocks that you analyzed.

$$\hat{R}_{i,t} = \hat{\beta}_0 + \hat{\beta}_1 \cdot R_{m,t}$$

```
reg_data_wide <- reg_data_wide |>
  mutate(
    BPI_Fitted = fitted(regression_results[["BPI.PS"]]),
    CNPF_Fitted = fitted(regression_results[["CNPF.PS"]]),
    MER_Fitted = fitted(regression_results[["MER.PS"]]),
    ABS_Fitted = fitted(regression_results[["ABS.PS"]]),
    TEL_Fitted = fitted(regression_results[["TEL.PS"]])
  )
```

4.2 Question 2 (5 pts)

Compute the residuals for each stock.

$$\hat{u}_{i,t} = R_{i,t} - \hat{R}_{i,t}$$

```
reg_data_wide <- reg_data_wide |>
  mutate(
    BPI_Residual = BPI.PS - BPI_Fitted,
    CNPF_Residual = CNPF.PS - CNPF_Fitted,
    MER_Residual = MER.PS - MER_Fitted,
    ABS_Residual = ABS.PS - ABS_Fitted,
    TEL_Residual = TEL.PS - TEL_Fitted
  )
```

4.3 Question 3 (3 pts)

What did you notice about the sum of the residuals for each stock?

```

residual_sums <- reg_data_wide |>
  summarise(
    BPI_Sum = sum(BPI_Residual, na.rm = TRUE),
    CNPF_Sum = sum(CNPF_Residual, na.rm = TRUE),
    MER_Sum = sum(MER_Residual, na.rm = TRUE),
    ABS_Sum = sum(ABS_Residual, na.rm = TRUE),
    TEL_Sum = sum(TEL_Residual, na.rm = TRUE)
  )

knitr::kable(t(residual_sums), col.names = c("Sum of Residuals"),
  digits = 14, caption = "Sum of Residuals by Stock")

```

Table 4: Sum of Residuals by Stock

	Sum of Residuals
BPI_Sum	0.000000000000000
CNPF_Sum	-0.000000000000001
MER_Sum	0.000000000000000
ABS_Sum	-0.000000000000007
TEL_Sum	0.000000000000003

Observation: The sum of residuals for every stock is practically zero (e.g., -0.000000000000007). This is a fundamental mathematical property of Ordinary Least Squares (OLS) regression when an intercept (β_0) is included. The positive and negative errors perfectly cancel each other out, leaving only a microscopic remainder due to the computer's floating-point precision limits.

5 Data Export

```

output_csv <- "H3_Group4_Regression_Output.csv"
write.csv(reg_data_wide, file = output_csv, row.names = FALSE)

```

The final dataframe has been successfully exported to `H3_Group4_Regression_Output.csv` containing **727 rows** and **17 columns** (Date, PSEi_Return, 5 stock returns, 5 fitted values, 5 residuals).